Lab Report – EE-170L Computer Programming

NAME	Danyal Hassan
ID	750
SEMESTER	2nd
LAB	Programming
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COMPUTER PROGRAMMING

Electrical Engineering Department — UET Nowshera

LAB 2

Lab Title: Basic Syntax and Input/Output in C++

1. OBJECTIVE

The aim of this lab is to develop an understanding of the fundamental structure of a C++ program. Students explore the use of input and output statements, practice working with various data types, apply correct indentation, and carry out calculations using arithmetic operators.

2. THEORY

Every C++ program is organized around a defined structure consisting of header file inclusions, a main function, and executable statements. The iostream library provides the tools needed for input and output operations. The cout statement sends data to the screen, while cin receives input typed by the user. The << and >> symbols serve as stream insertion and extraction operators respectively.

Since C++ is a statically typed language, each variable must be explicitly declared with its data type before being used in the program. Commonly used data types include:

- int — for integer values
- double — for decimal (floating-point) numbers
- char — for single characters
- bool — for true/false values
- string — for sequences of text

C++ also provides arithmetic operators such as +, −, *, /, and % for mathematical calculations. Proper indentation makes programs easier to read. Comments are written using // for single-line comments or /* */ for multi-line comments.

3. SOFTWARE USED

- DEV-C++ IDE
- C++ Compiler (bundled with DEV-C++)

4. PROGRAMS

Program 1 — Hello World and User Greeting

This program prompts the user to provide their name and then prints a personalised welcome message on the screen.

Algorithm:

- Start the program.
- Declare a string variable to store the user's name.
- Prompt the user to enter their name.
- Read the name using cin.
- Display a greeting message using cout.
- End the program.

```
// Include iostream for input/output operations
#include <iostream> using
namespace std;

int main()
{
    string name;

    // Prompt the user to enter their name
cout << "Enter your name: ";      cin >>
name;

    // Display a personalised greeting
    cout << "Hello " << name << "! Welcome to C++ programming.";

    return 0; // Program ends successfully
}
```

Program 2 — Simple Calculator Using Arithmetic Operators

This program takes two integer values from the user and computes addition, subtraction, multiplication, and division, printing each result to the console.

Algorithm:

- Start the program.
- Declare two integer variables.
- Take two numbers as input from the user.
- Perform addition, subtraction, multiplication, and division.
- Display each result using cout.
- End the program.

```
// Include iostream for input/output operations
#include <iostream> using
namespace std;

int main()
{
    int a, b; // Declare two integer variables

    cout << "Enter two numbers: ";
    cin >> a >> b; // Read both numbers

    // Perform and display each arithmetic operation
cout << "Addition       = " << a + b << endl;    cout
<< "Subtraction    = " << a - b << endl;     cout <<
"Multiplication = " << a * b << endl;      cout <<
"Division       = " << a / b << endl;

    return 0;
}
```

5. CONCLUSION

Throughout this lab, the fundamental structure of a C++ program was studied and put into practice. The use of cin and cout for handling input and output was demonstrated, various data types were examined, and arithmetic operators were applied to carry out calculations. Mastering these basics is essential for building any C++ application.